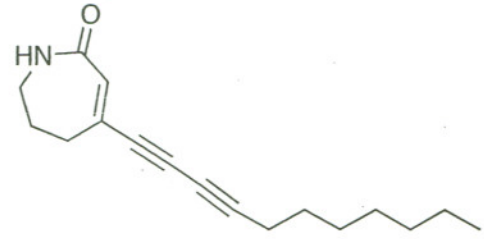
Organic Cume
February 6, 2004

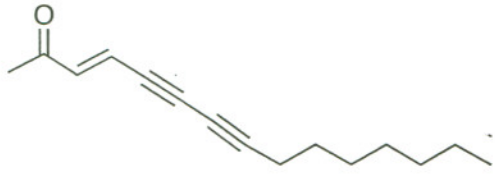
The following diynes, designated as montiporynes, were isolated from a stony coral (*Montipora* sp.). Match the structures of montiporynes A-F with the ^{1}H NMR data sets. Briefly indicate what criteria you used to distinguish each one from the others, i.e. what signals are unique to each compound?

A

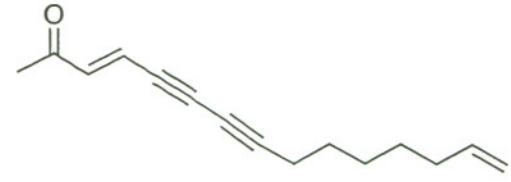


B

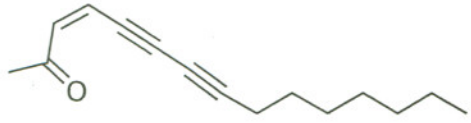
C



D



E



F

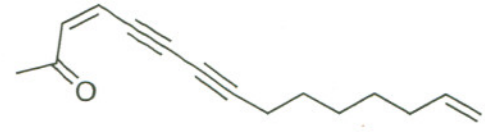


Table 2. ^{1}H NMR Data of **1**–**4**[a] (300 MHz, CD_3OD)

1	2	3	4
2.25 (s)	2.38 (s)	2.25 (s)	2.38 (s)
6.57 (d, 16.1)	6.28 (d, 11.5)	6.55 (d, 16.1)	6.28 (d, 11.5)
6.73 (d, 16.1)	6.46 (d, 11.5)	6.72 (d, 16.1)	6.46 (d, 11.5)
2.39 (t, 6.9)	2.39 (t, 6.9)	2.38 (t, 6.9)	2.38 (t, 6.9)
1.56 (quint, 7.4)	1.56 (quint, 7.3)	1.56 (quint, 7.2)	1.56 (quint, 7.2)
1.42 (quint, 5.2)	1.29–1.42 (m)	1.28–1.42 (m)	1.28–1.42 (m)
1.29–1.35 (m)	1.29–1.42 (m)	1.28–1.42 (m)	1.28–1.42 (m)
0.91 (t, 7.0)	0.90 (t, 7.0)	2.10 (quart, 6.8)	2.10 (quart, 6.8)
		5.80 (ddt, 17.0, 10.2, 6.8)	5.80 (ddt, 17.0, 10.2, 6.8)
		4.91 (dd, 10.2, 2.1)	4.91 (dd, 10.2, 2.1)
		4.98 (dd, 17.0, 2.1)	4.98 (dd, 17.0, 2.1)

Table 3. ^{13}C NMR Data of **1**–**4**[a] (CD_3OD)

1	2	3	4
27.5	27.5	27.5	27.5
199.3	199.3	199.3	199.3
141.2	140.6	141.2	140.5
124.3	120.5	124.2	120.5
65.5[b]	65.6[b]	65.5[b]	65.5[b]
72.6[b]	72.7[b]	72.7[b]	72.7[b]
84.9[b]	84.9[b]	84.7[b]	84.7[b]
90.5[b]	90.5[b]	90.5[b]	90.5[b]
20.2	20.2	20.2	20.2
29.2	29.2	29.1	29.1
29.9, 29.8	29.9, 29.8	29.8–30.7	29.8–30.7
32.8	32.8	29.8–30.7	29.8–30.7
23.6	23.6	29.8–30.7	29.8–30.7
14.4	14.4	34.8	34.8
		140.1	140.1
		114.7	114.7

5		6	
^{1}H	^{13}C	^{1}H	^{13}C
			189.5
	170.5	5.89 (s)	126.5
6.60 (s)	115.7		164.2
	144.8	2.62 (dd, 16.7, 4.6)	37.6
		2.49 (dd, 16.7, 9.0)	
2.75 (td, 6.5, 2.0)	28.2	3.19 (tt, 9.0, 4.6)	28.7
1.86 (quint, 5.4)	23.4	2.57 (dd, 16.4, 4.6)	43.4
		2.42 (dd, 16.4, 9.0)	
3.35 (t, 5.7)	42.9	2.00 (s)	24.5
	65.7[b]		65.9[b]
	72.5[b]		68.3[b]
	85.5[b]		78.8[b]
	90.3[b]		80.0[b]
2.39 (t, 6.6)	20.2	2.25 (t, 6.6)	19.7
1.56 (quint, 7.4)	29.3	1.50 (quint, 7.3)	29.4
1.42 (quint, 7.5)	29.9[c]	1.39 (quint, 6.9)	29.9[c]
1.29–1.34 (m)	29.8[c]	1.29–1.35 (m)	29.8[c]
1.29–1.34 (m)	32.7	1.29–1.35 (m)	32.9
1.29–1.34 (m)	23.6	1.29–1.35 (m)	23.7
0.91 (t, 7.1)	14.4	0.90 (t, 7.0)	14.6